

Module : 101-107

Consider X with a metric d . For any $x \in X$ and any real number $r > 0$, the set of all points $y \in X$ whose distance from x is less than r is called open ball of radius r and centered at x i.e. $B(x,r) = \{ y \in X \mid d(x,y) < r \}$

Prepare Page # 33-34-35

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Metric Topology: Let X be a non empty set with metric d . The topology T on X generated by the set of all open balls in X with respect to d is called metric topology.

A topology T on X induced by metric d .

Consider X with a metric d , this d induces a topology "metric topology". (X,d) is called **metric space**.

The topology generated by the set of all singletons is discrete topology.

Module : 108-110

Metrizable Space: A topological space (X,T) is called Metrizable space if there exists a metric d on X that induces topology T .

Module : 108-110

Metrizable Space: A topological space (X, T) is called Metrizable space if there exists a metric d on X that induces topology T .

Not all spaces are Metrizable.

Let X be an infinite set. Consider cofinite topology on X then it is not Metrizable space.

$\mathbb{R}$ with lower limit topology is not Metrizable space.

Module : 111-112

A topological space (X, T) is called a **First Countable Space** if for each $p \in X$ there exists a countable local base at p .

Every subspace of first countable space is first countable.

Discrete space is first countable space.

Every finite set with any topology is first countable.

$\mathbb{R}$ with usual topology is first countable i.e $(\mathbb{R}, T_u)$ is first countable.

Every metric space is first countable.

Module : 113-114

A topological space (X, T) is called **Second Countable Space** if there exist a countable a countable basis for T .

Every subspace of a second countable space is second countable.

Let X be a countable set with discrete topology then (X, T_{dis}) is second countable.

Any finite set with any topology is second countable.

Every subspace of a second countable space is second countable.

Let X be a countable set with discrete topology then (X, T_{dis}) is second countable.

Any finite set with any topology is second countable.

$\mathbb{R}$ with usual topology is second countable i.e. $(\mathbb{R}, T_u)$ is second countable.

Module : 115-116

Second countability implies first countability.

Let a topological space (X, T) is second countable space then it is also first countable space.

First countability does not imply second countability.

Consider $\mathbb{R}$ with usual topology then it is both first countable and second countable.

Consider $\mathbb{R}$ with lower limit topology then it is first countable but not second countable.

A first countable space may or may not be second countable.

Module : 117-119

Cover: A cover of a set X is a collection of sets $U = \{U_i \mid i \in I\}$ where I is index set such that $X \subset \bigcup_{i \in I} U_i$

Cover: Let (X, T) be topological space. A cover of a X is a collection of subsets of X i.e. $U = \{U_i \mid U_i \subset X \mid i \in I\}$ where I is index set such that $X = \bigcup_{i \in I} U_i$

Open Cover: Let (X, T) be topological space. An open cover of a X is a collection of subsets of X i.e.

$U = \{U_i \mid U_i \in T \mid i \in I\}$ where I is index set such that $X = \bigcup_{i \in I} U_i$

Module : 120-121

COLLECTION BY ABU SULTAN **** MTH634 **** DATA FOR FINAL TERM **** Page 2 of 9

Lindelof Space: A topological space (X, T) is called a Lindelof space iff every open cover of X has a countable sub cover.

Let X be a finite set with any topology then it is Lindelof.

A set with indiscrete topology then it is Lindelof.

$X = \mathbb{R}$ with usual topology is Lindelof.

A subspace of Lindelof space needs not to be Lindelof.

Every countable space is second Lindelof. (ALSO PREPARE ITS PROOF FROM P#69-70)

Module : 122-125

A topological space (X, T) is said to be separable if there exist a countable dense subset A of X .

Every second countable space is separable.

$\mathbb{R}$ with lower limit topology is separable but not second countable.

Let X be a finite set then X with any topology is separable.

Consider a set X with indiscrete topology the X is separable.

$\mathbb{R}$ with usual topology is separable.

$\mathbb{R}$ with discrete topology is not separable.

X is separable iff X is countable.	
(X, T_{cof}) is separable.	[ALSO PROOF FROM PAGE # 74-75]
In general subspace of separable is not separable.	
Module : 126	
A metric space is not a second countable and not separable in general.	
An infinite set X with trivial metric is not second countable and not separable.	
A separable metric space is second countable.	[ALSO PROOF FROM PAGE # 78]
Module : 127-143	
There are 7 separation axioms. T_0 -Spaces, T_1 -Spaces, T_2 -Spaces (Hausdorff), T_3 -Spaces, Normal Spaces, T_4 -Spaces,	
<u>T_0-Space:</u> A topological space (X, T) is said to be T_0-Space iff for each $x, y \in X$ there exist an open set $U_x \subset X$ containing x such that $y \notin U_x$ OR There exist an open set $V_y \subset X$ containing y such that $x \notin V_y$.	
T_0 Property is a topological property.	
<u>T_1-Space:</u> A topological space (X, T) is said to be T_1-Space iff for each $x, y \in X$ such that $x \neq y$ there exist open sets U_x, U_y of X containing x, y respectively such that $y \notin U_x$ and $x \notin U_y$.	
T_1 -Property is a topology property.	

Every discrete space is T_1 space.

Every metric space is T_1 space.

Every cofinite space is T_1 space.

T_1 property is much stronger property than T_0 property.

A topological space (X, T) is T_1 iff every singleton subset $\{x\} \subset X$ is closed. [ALSO PROOF FROM PAGE # 90]

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Every subspace of a T_1 space is also a T_1 space. [ALSO PROOF FROM PAGE # 92]

Every finite subset of T_1 space is closed.

Let A be a finite subset of a T_1 space X then $\bar{A}$ is empty.

T_2 -Space: A topological space (X, T) is said to be **T_2 -Space** iff for every distinct pair of element $x, y \in X$ there exist open sets U_x, U_y of X containing x, y respectively such that $U_x \cap U_y = \emptyset$.

T_2 -space is also called Hausdorff discovered by Felix Hausdorff (1868-1942)

T_2 property is a topological property.

Every discrete space is Hausdorff.

A nonempty set $X \neq \{x\}$ with indiscrete topology is not Hausdorff.

$\mathbb{R}$ with lower limit topology is Hausdorff.

$\mathbb{R}$ with lower limit topology is Hausdorff .
$\mathbb{R}$ with topology generated $\{(a, \infty) a \in \mathbb{R}\}$ is not Hausdorff.
An infinite set with cofinite topology is not Hausdorff space.
The property of being a Hausdorff space is inherently.
Let (X,T) is a Hausdorff space and $A \subset X$ then (A,T_A) is Hausdorff. [ALSO PROOF FROM PAGE # 102]
An infinite set X with cofinite topology is not Metrizable.
Every metric space is Hausdorff. [ALSO PROOF FROM PAGE # 104]
T_2 property is a stronger property than T_1 property.
Every T_2 space is T_1 space but every T_1 space is not T_2 space.
An infinite set X with cofinite topology is T_1 space but not T_2 space.
Let X be a Hausdorff space then every convergent sequence of points has a unique limit point.
Module : 144
A topological space (X,T) is said to be regular iff for every closed subset F of X and for every $x \in F^c \exists$ open subsets U_F and U_x containing F and x such that $U_F \cap U_x = \phi$
A topological space (X,T) is said to be " T_3 space" iff it is either Regular+ T_0 or Regular+ T_1 or Regular+ T_2 .
Every T_3 space will be regular but every regular space will not T_3 space.
Property of being regular is a topological property.
Every subspace of regular space is a regular space.

Every subspace of regular space is a regular space.

Every subspace of a T_1 space is also a T_1 space i.e. T_1 space is heredity.

T_3 property is a topological property.

Every subspace of a T_3 space is a T_3 space.

Module : 145

Discrete space is a regular space.

Indiscrete space is a regular space.

A discrete topology space has a complete power set.

Every subset of discrete space is also closed as well as open so open collection is also close collection.

$\mathbb{R}$ with usual property is regular and this space is also T_3 space.

An infinite set with co finite topology is not regular so it is not T_3 .

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K-Topology on real line topology $T_K, \dots, (\mathbb{R}, T_k)$ is not a regular space.

Module : 146

A topological space is said to be normal iff for every pair of disjoint closed subset $F_1, F_2 \subset X$, there exist open subset U_{F_1} and U_{F_2} containing F_1 and F_2 respectively such that $U_{F_1} \cap U_{F_2} = \phi$

A topological space (X, T) is said to be T_4 space iff it is either Normal + T_1 or Normal + T_2

Module : 146

A topological space is said to be normal iff for every pair of disjoint closed subset $F_1, F_2 \subset X$, there exist open subset U_{F_1} and U_{F_2} containing F_1 and F_2 respectively such that $U_{F_1} \cap U_{F_2} = \phi$

A topological space (X, T) is said to be T_4 space iff it is either Normal + T_1 or Normal + T_2

Property of being normal is a topology property.

Subspace of normal space needs not to be normal space.

T_4 space is a topology property.

A topological space (X, T) is said to be complete normal space iff every subspace of X with subspace topology is normal.

Module : 147

Consider: $x = \{a, b, c, d\}$

$T = \{ \phi, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X \}$

Claim: (X, T) is normal space but (X, T) is not T_4 .

Consider: $y = \{b, c, d\}$

$T = \{ \phi, \{d\}, \{b, d\}, \{c, d\}, X \}$

Claim: (y, T_4) is not a normal space.

Consider: $x = \{a, b, c, d\}$

$T = \{ \phi, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X \}$

Claim: (X, T_4) is not completely normal space.

Question # 2 of 10 (Start time: 05:30:56 AM, 27 August 2022)

Total Marks

Metric topology induced by $d(x, y) = |x - y|$ on $\mathbb{R}$ is called ______.

Select the correct option

[Reload Math Equation](#)

- | | |
|----------------------------------|---------------------|
| ☐ | None of them |
| ☐ | discrete topology |
| ☐ | indiscrete topology |
| ☒ | usual topology |



Question # 1 of 10 (Start time: 05:30:08 AM, 27 August 2022)

Total M

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{3\}, \{4\}, \{3, 4\}, X\}$ be a topology on X , then which of the following is true?

Select the correct option

Reload Math Equat

- ☒ All of them
- ☐ The set $\{\emptyset, \{3\}, \{4\}\}$ is an open sub-cover of $\{\emptyset, \{3\}, \{4\}, X\}$.
- ☐ The set $\{\emptyset, \{3\}, \{4\}\}$ is an open cover of the set $\{4\}$.
- ☐ The set $\{\emptyset, \{3\}, \{4\}, X\}$ is an open cover of the set $\{4\}$.



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DISUBSCRIBE



A map f from a topological space X to a topological space Y is called a closed map if and only if the image of each closed subset of X is _____ in Y .

Select the correct option

☐	open
☐	contained
☒	closed
☐	empty



Question # 8 of 10 (Start time: 05:36:44 AM, 27 August 2022)

1

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 1$ is.....

Select the correct option

 Reload Math

- | | |
|----------------------------------|---------------------------------|
| ☒ | $\{\{1\}, \{1, 2\}, X\}$ |
| ☐ | $\{\{1\}, \{2\}, X\}$ |
| ☐ | $\{\{1\}, \{2\}, \{1, 2\}, X\}$ |
| ☐ | None of them |



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MTH634 - Topology (Quiz No. 3)

Quiz Start Time

Question # 6 of 10 (Start time: 05:34:58 AM, 27 August 2022)

Homomorphism of topological spaces is an equivalence relation.

Select the correct option

False

☐

True

☒

Question # 10 of 10 (Start time: 05:38:17 AM, 27 August 2022)

Let X and Y be topological spaces and $f : X \rightarrow Y$ be a bijection then which of the following statement is NOT true?

Select the correct option

- ☐ f is discontinuous and open.
- ☐ f is homeomorphism.
- ☐ f is continuous and closed.
- ☐ None of them



Question # 4 of 10 (Start time: 05:32:48 AM, 27 August 2022)

To

Which of the following statement is true?

Select the correct option

- | | |
|-----------------------|--|
| ☐ | Restriction of a closed map may or may not be an open map. |
| ☐ | Restriction of an open map can never be an open map. |
| ☐ | Restriction of an open map may or may not be an open map. |
| ☐ | Restriction of an open map is always an open map. |

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Question # 9 of 10 (Start time: 09:33:37 AM, 27 August 2022)

A map f from a topological space X to a topological space Y isif and only if the image of each closed subset of X is closed in Y .

Select the correct option

- | | |
|-----------------------|------------------|
| ☐ | a closed map |
| ☐ | an injective map |
| ☐ | an open map |
| ☐ | a surjective map |



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MTH634 - Topology (Quiz No. 3)

Quiz Start Time: 0

Question # 7 of 10 (Start time: 05:35:44 AM, 27 August 2022)

The property $d(x, y) = 0$ of a metric space is called-----

Select the correct option

- | | |
|-----------------------|--------------------|
| ☐ | Triangle property |
| ☐ | Reflexive property |
| ☐ | Non negativity |
| ☐ | Symmetric property |



B

